BIS 527

Project proposal presentations

Thursday, October 1, 2026 · Five minutes per team · Presentation order

1. 404 Sleep Not Found
2. Five Shocking Worms
3. ShuaQi Team
4. Biostatisticats
5. DatumDance
6. O(n log n)
7. DataPulse
8. MindMerge
9. Pierpont
10. jet2holiday
11. M4zi1

PRESENTATION 1 OF 11

404 Sleep Not Found

Next: Five Shocking Worms

BIS 527 · Introduction to Health Data Science

Predicting District-Level Dengue Incidence in Peru

A comparative machine learning and statistical approach

Team

404 Sleep Not Found

eja45	Edson Ascencio
fy94	Fuyang Yu
jl3954	Nicole Luo
qq27	Elaine Qin
yq257	Yifei Qu

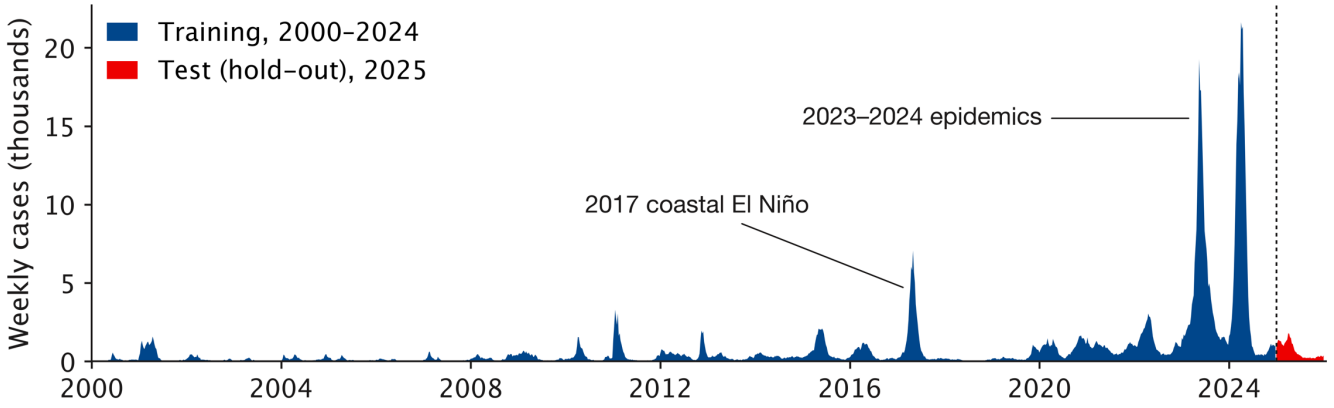
Proposal presentation · Thursday, October 1, 2026

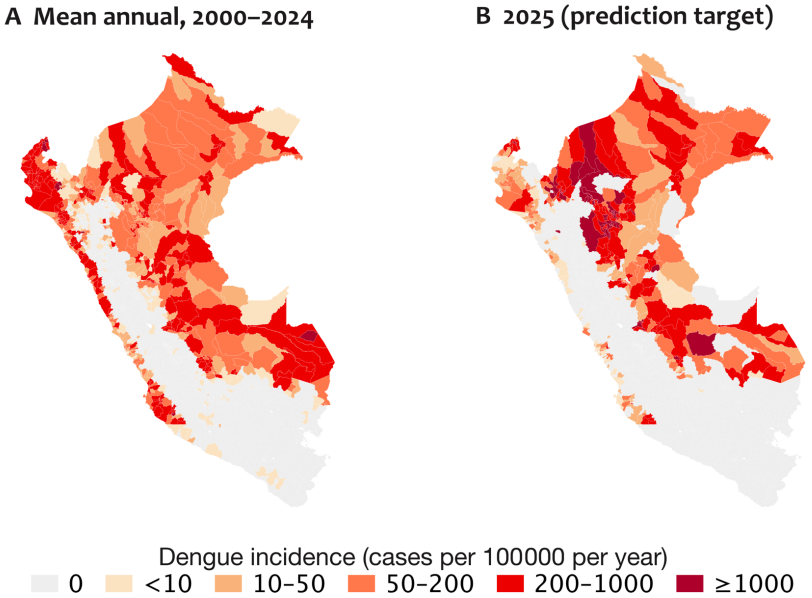


Background and main question

Dengue in Peru is rising, uneven across districts and hard to anticipate

Dengue is an important public health concern in Peru. Accurate local-level predictions could help identify high-risk areas and support targeted prevention and resource allocation.





MAIN QUESTION

Primary Among Random Forest, SVM, XGBoost and zero-inflated negative binomial (ZINB) regression, trained on 2000–2024 data, which model most accurately predicts 2025 dengue incidence at the district (ADMIN3) level?

Secondary Which variables (temperature, NDVI, ENSO, population, historical dengue) matter most for prediction?

1891
districts

1357
epidemiological weeks

2·57 M
district-weeks

Resources

One public district-week panel; how its cases fall across seasons and regions

Dengue cases
CDC–MINSA surveillance [1,2]

Population
INEI projections, WorldPop [3,4]

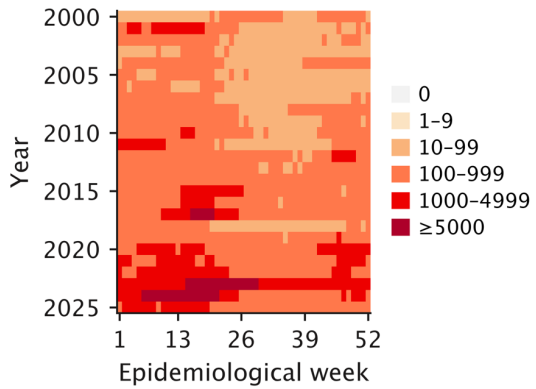
Climate
ERA5–Land, CHIRPS, MODIS LST [5–7]

Vegetation and ENSO
MODIS NDVI/EVI, NOAA ONI and Niño SST [8,9]

Terrain and boundaries
SRTM, INEI districts and regions [10–12]

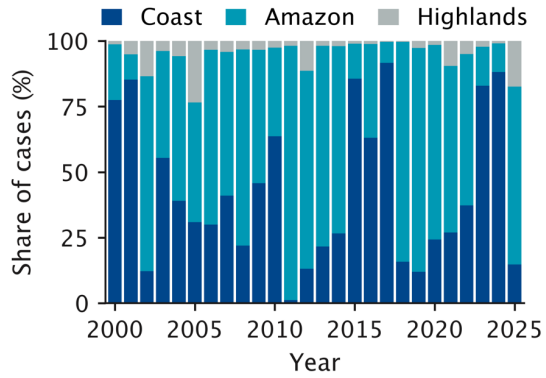
District-week panel
1891 districts, 1357 weeks (2000–2025); gridded data averaged per district in Google Earth Engine [13]
Train 2000–2024: 1 029 421 cases
Test 2025: 36 120 cases

A Weekly cases by year



Cases concentrate in weeks 10–22 (March to May) every year; 2023 and 2024 stand out.

B Cases by natural region



The coast drove 2023–2024; the Amazon led most other years, including the 2025 test year.

Access Full access; every source is public. No personal identifiers: cases are aggregated by district and week.

To note:
Temperature, NDVI and ENSO may affect transmission with a delay rather than immediately. We will compare predictors at different lag periods and keep a lag only if it improves out-of-sample performance, after aligning every source to one weekly resolution.

Relevant literature

[5] Muñoz-Sabater et al. 2021, ERA5-Land, doi:10.5194/essd-13-4349-2021
[6] Funk et al. 2015, CHIRPS, doi:10.1038/sdata.2015.66
[7,8] Wan et al. 2021; Didan 2021, MODIS land surface temperature, NDVI
[13] Gorelick et al. 2017, Google Earth Engine, doi:10.1016/j.rse.2017.06.031

Project path and timeline

Proposed approach, then tentative dates and expected outputs

Panel
District-week cases, population, climate, ENSO, terrain

→

Features
Harmonize and standardize predictors

→

Baseline
OLS on historical dengue incidence

→

Models
Random Forest
XGBoost · SVM
ZINB

→

Evaluate on 2025
RMSE and RSE out of sample; variable importance

Why these models RF, XGBoost and SVM capture non-linear climate-dengue relationships; ZINB handles the excess of zero weeks; OLS is the benchmark that every model must beat.

Oct 1

Proposal

Finalise research question, verify access to datasets, inspect initial data structure.

Oct 10

First analysis

Clean and merge spatio-temporal data, data wrangling, build an OLS baseline model.

Oct 29 tentative

Midpoint check-in and presentation

Train initial predictive models, report preliminary metrics from cross-validation.

Nov 16

Final results

Feature importance ranking, hyperparameter optimisation, model comparison, visualisation, final poster and report.

PRESENTATION 2 OF 11

Five Shocking Worms

Next: ShuaQi Team

Hospital Financial Distress

Medicare exit prediction and investment risk

Five Shocking Worms



Team members

BIS 527 project team

NetID	Full name
sr2696	Shuolin Ren
ll2359	Liangbang Li
xs354	Xinyu Sheng
zw535	Zirun Wang
yl2995	Yiqi Liu

Research group update
October 1, 2026

Background and research questions

Financial distress may precede Medicare exit, while investment income can buffer operating losses.

Prediction

Can financial statements predict Medicare exit 1–3 years ahead?

Investment exposure

Did hospitals with larger portfolios experience greater FY2022 losses?

Evidence informing the design

Holmes, Kaufman, and Pink (2017)

Two-year rural distress index. Validation c-statistic: 0.74 for financial distress.

Kaufman et al. (2016)

Lower profitability, liquidity, and equity preceded rural hospital closures.

Altman (1968) and Shumway (2001)

Combine accounting ratios and model the timing of financial failure.

Contribution: national acute-care data, explicit exit definitions, and fiscal-year market exposure.

Key literature: JRural Health, 2017 and 2016; Journal of Finance, 1968; Journal of Business, 2001.

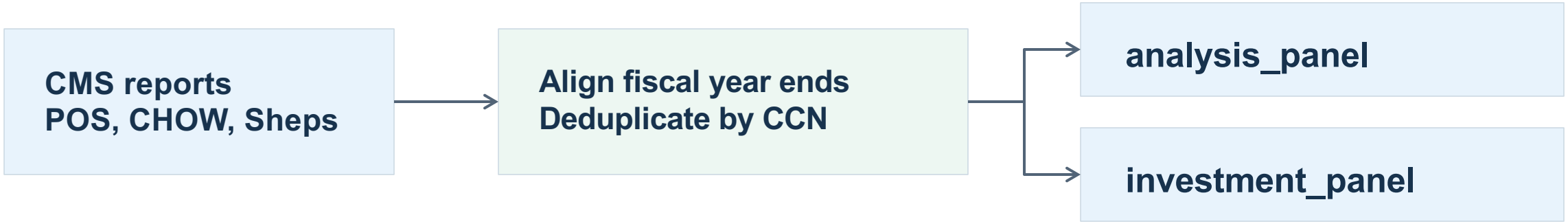
Study population and data pipeline

3,820
hospitals

42,383
hospital-years

766
linked Medicare exits

Current descriptive subset: fiscal year ends in 2011–2023. Exit follow-up through June 2026.



Core predictors

Profitability and liquidity
Leverage and payer mix
Size, ownership, and rurality

Data and label safeguards

Six-digit CCNs and 1st/99th percentile winsorization
Receipt date or fiscal year end + 5 months
Broad exit and acquisition-excluded labels

Sources: local CMS reports and POS Q2 2026. Descriptive counts differ from the extended proposal panel.

Prediction design and evaluation

Target: Medicare exit within 1, 2, or 3 years after a report becomes available

2011–2018
Training

2019–2021
Tuning

2022–2024
Testing*

Baseline model	Inputs and purpose
Regularized logistic regression	Core financial ratios + ownership + rurality
Discrete-time hazard (cloglog)	Core predictors + baseline time terms
Operating-margin-only logistic	Benchmark for the value of additional predictors

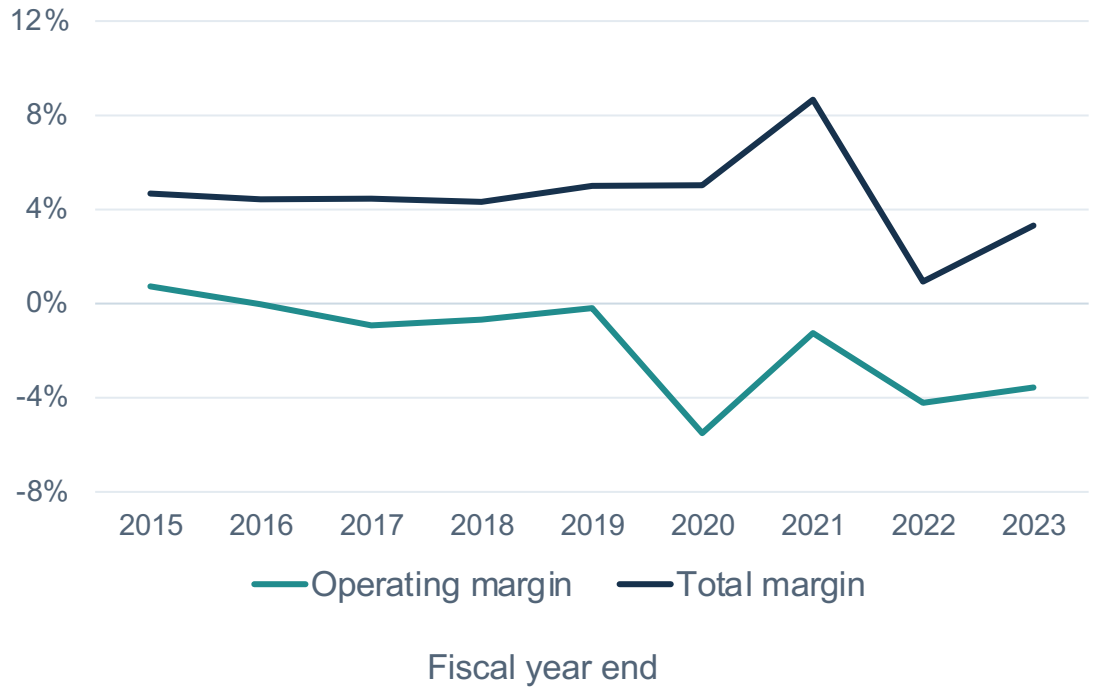
Evaluation PR-AUC, Brier score, KS, calibration, and coefficient direction

*** Require availability date + horizon ≤ June 30, 2026. Test eligibility differs by horizon.**

Planned baseline analyses. No model performance estimates are available yet.

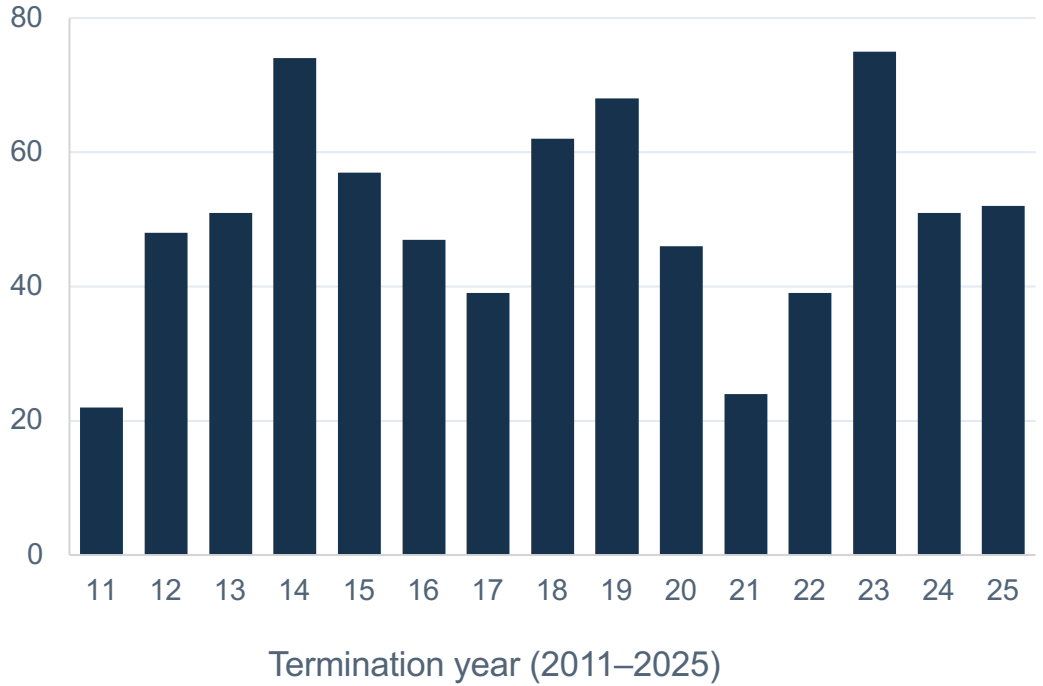
Preliminary financial trends and exit labels

Annual median financial margins



FY2022 median: −4.2% operating, +0.9% total

Medicare exits among linked hospitals



527 code-01 exits mix mergers and closures

Local CMS/POS calculations. Annual medians are descriptive. 2026 includes 11 exits through June and is omitted from the chart.

Investment exposure and the FY2022 shock

Hypothesized financial pathway

Stock and bond market losses → **Asset values fall**
Other income falls → **Total margin and liquidity weaken** → **Greater financial distress**

Two-way fixed-effects difference-in-differences

$$Y_{it} = \alpha_i + \lambda_t + \beta(Heavy_i \times FY2022_t) + \varepsilon_{it}$$

Comparison and outcomes

2019 top quartile vs. bottom half of investment share
Other income / NPR, investment growth,
total margin, and days cash on hand

Identification and market alignment

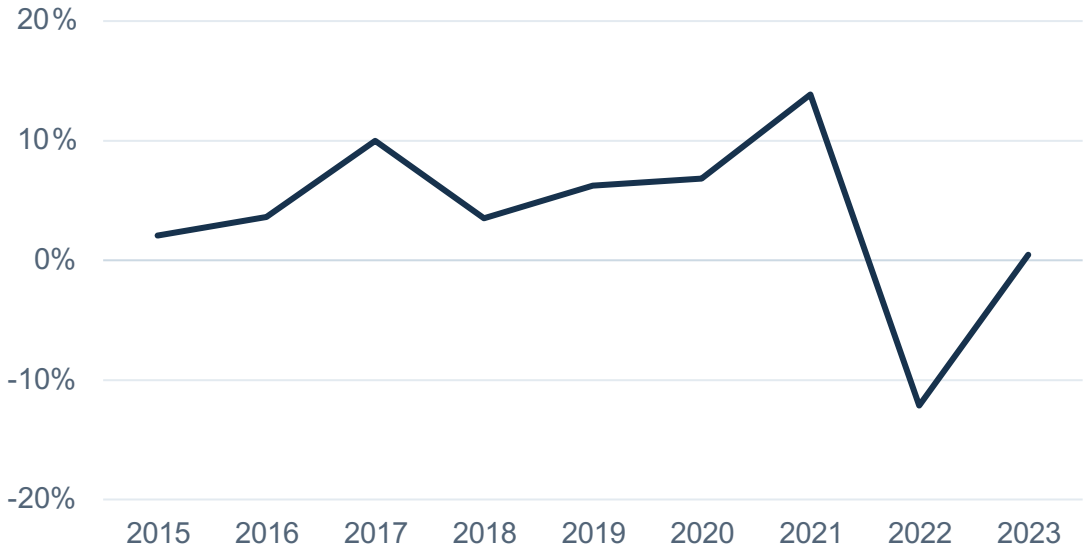
FY2015–2024; exclude 2020–2021 in the main model
Hospital-clustered standard errors; assess pretrends
Compound monthly stock and bond returns by fiscal year

Planned analysis. Also estimate Heavy × Post2022. DGS10 is a yield series, not a bond total-return series.

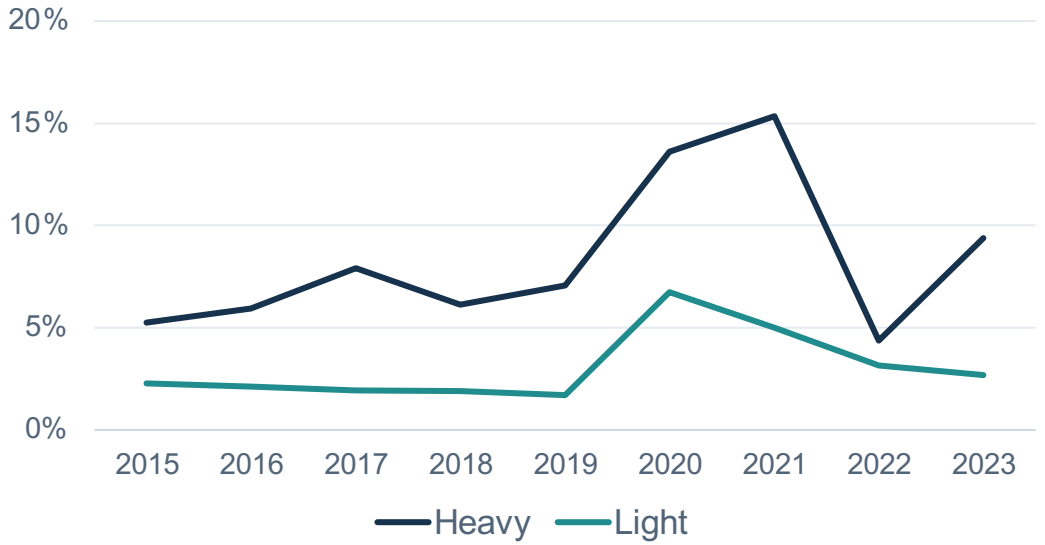
Preliminary investment trends

Groups fixed using 2019 investment share: heavy ≥16.2% (n=791), light ≤0.18% (n=1,582)

Median annual investment-balance growth



Median other income / net patient revenue



FY2022 investment growth: −12.1% in heavy hospitals. DiD estimates are pending.

Growth requires consecutive years and prior holdings >$1 million. Other income includes relief funds in 2020–2021.

Midpoint deliverables and outlook

Target: both baseline modules ready for the tentative October 19 check-in

Owner	Deliverables before midpoint	Due
Xinyu Sheng	Source audit; monthly markets; two analysis panels	Oct 5 / 9 / 10
Yiqi Liu	Literature + codebook; labels; three descriptive figures	Oct 5 / 7 / 14
Shuolin Ren	Prediction-panel acceptance; three baseline models	Oct 10 / 15
Liangbang Li	Investment-panel acceptance; fixed-effects DiD	Oct 10 / 15
Zirun Wang	Shared figure style; integrated slides and report	Oct 17

Current evidence

Financial and investment patterns warrant testing. Accurate labels and reporting dates remain essential.

Next steps

Resolve labels and FY2024 fields. Evaluate baselines, calibration, and pretrends before drawing causal conclusions.

All members submit results and brief interpretations by Oct 15. Final integration Oct 17; midpoint date remains tentative.

PRESENTATION 3 OF 11

ShuaQi Team

Next: Biostatisticats

BIS 527 PROJECT PROPOSAL

Sleep Duration and Depressive Symptoms

Among U.S. young adults ages 18-35

Research question

Is shorter sleep associated with greater depressive symptom severity?

Team Shuaqi

Richard Zhu · Muyuan Zhu · Yiang Luo · Yunjia Huang



Research question and study framework

Question

Does shorter usual sleep duration correspond to more severe depressive symptoms in U.S. young adults?

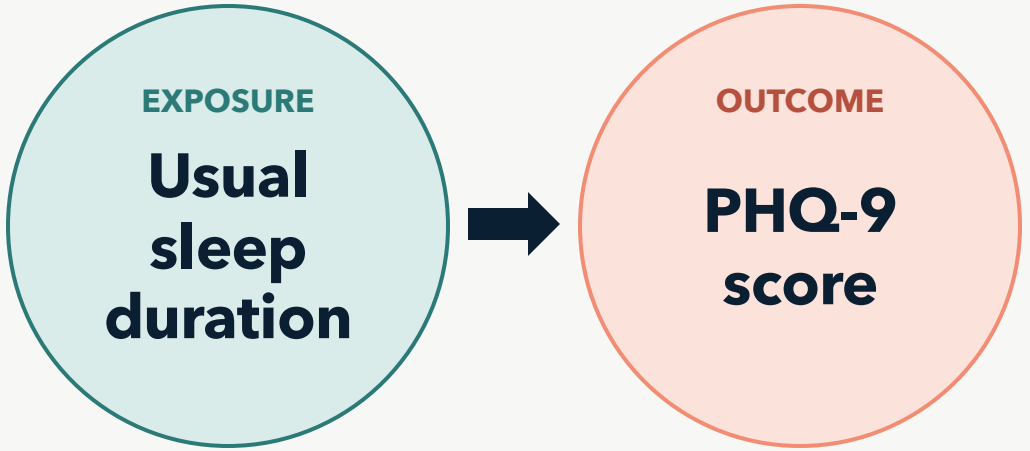
Population

Adults ages 18-35 in NHANES 2021-2023

Purpose

Test whether the association remains after accounting for demographic and health-related factors.

Conceptual model



Adjustment factors

Age · Sex · Race/ethnicity · Household income
BMI and physical activity may be added

Cross-sectional data support association, not causation

Dataset and analysis plan

NHANES 2021-2023

Public CDC/NCHS data merged by participant identifier SEQN

1. **Sleep Disorders Questionnaire**
   Usual sleep duration
2. **Depression Screener**
   Nine PHQ-9 items
3. **Demographics**
   Age, sex, race/ethnicity, income
4. **Possible covariates**
   BMI and physical activity

Eligible sample

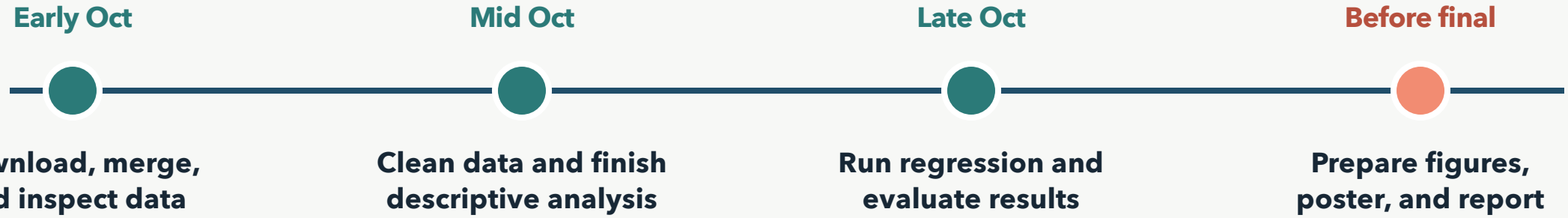
Ages 18-35 with available sleep and depression data

Analysis sequence

1. **Merge and restrict**
   Combine files, keep ages 18-35, inspect missing data
2. **Describe distributions**
   Summarize sleep duration and PHQ-9 outcomes
3. **Compare sleep groups**
   <6 h · 6 to <7 h · 7-9 h · ≥9 h
4. **Estimate associations**
   Regression with selected covariates
5. **Sensitivity analysis**
   Recalculate depression score without the sleep item

Work plan, uncertainty, and next step

Timeline



MAIN UNCERTAINTY

Will enough complete observations remain?

The team must also finalize how sleep duration and depressive symptoms will be defined.

IMMEDIATE NEXT STEP

Build the analytic sample

Download the sleep, depression, and demographic files; merge by SEQN; then determine the final eligible sample size.

All milestones are assigned to all team members

PRESENTATION 4 OF 11

Biostatisticats

Next: DatumDance

Who Pays After Medicaid Expansion?

A State-Level Analysis of the Payer Mix of U.S. Health Spending

Team name

Biostatisticats

Thursday, October 1, 2026

Team members

Sort alphanumerically by NetID.

NetID	Full name
gl645	Leon Liu
hx356	Daniel Xia
rm2896	Ruoyun Ma
yl2966	Yuhong Liu
zy372	Zhen Yang

Background and main question

Background

Medicaid expansion under the Affordable Care Act increased coverage for low-income adults, but its broader effects on who pays for healthcare remain less clear.

Main question

How did Medicaid expansion change the payer mix of state-level healthcare spending from 2010 to 2019?

Project path and timeline

Tentative dates and expected outputs

Oct 1

Proposal

Finalize question, datasets, and analysis scope.

Oct 15

Data preparation

Merge IHME, KFF, and CPS state-year data; produce descriptive spending trends.

Oct 29

Tentative

Midpoint check-in and presentation

Show baseline plots, initial DiD estimates, and pre-trend checks.

Nov 10

Model development

Compare payer-specific outcomes and test alternative treatment definitions.

Proposed approach

DiD with state/year fixed effects + RF/XGBoost for nonlinear and heterogeneous patterns

$$Y_{st} = \beta Expanded_{st} + \gamma X_{st} + \alpha_s + \lambda_t + \epsilon_{st} \quad + \quad \hat{Y}_{st} = f(X_{st}, Expanded_{st})$$

Resources

Dataset(s)

IHME. *United States health spending by state, payer type, and service, 2003–2019.* https://ghdx.healthdata.org/record/ihme-data/united-states-health-spending-by-state-payer-type-service-2003-2019

KFF. *State Activity Around Expanding Medicaid Under the Affordable Care Act.* https://www.kff.org/affordable-care-act/state-indicator/state-activity-around-expanding-medicaid-under-the-affordable-care-act/

U.S. Census Bureau. *Current Population Survey Annual Social and Economic Supplement.* https://www.census.gov/data/datasets/time-series/demo/cps/cps-asec.html

Relevant Literature

[1] Pines et al. 2016 — Medicaid expansion and emergency department payer mix

https://pubmed.ncbi.nlm.nih.gov/27503974/

Shows how DiD can be used to estimate Medicaid expansion–related shifts in payer mix.

[2] Nikpay et al. 2017 — Medicaid expansion, inpatient payer mix, and hospitalizations

https://pubmed.ncbi.nlm.nih.gov/28957347/

Distinguishes changes in payer composition from changes in total healthcare use.

PRESENTATION 5 OF 11

DatumDance

Next: O(n log n)

Wandering Safely in NYC

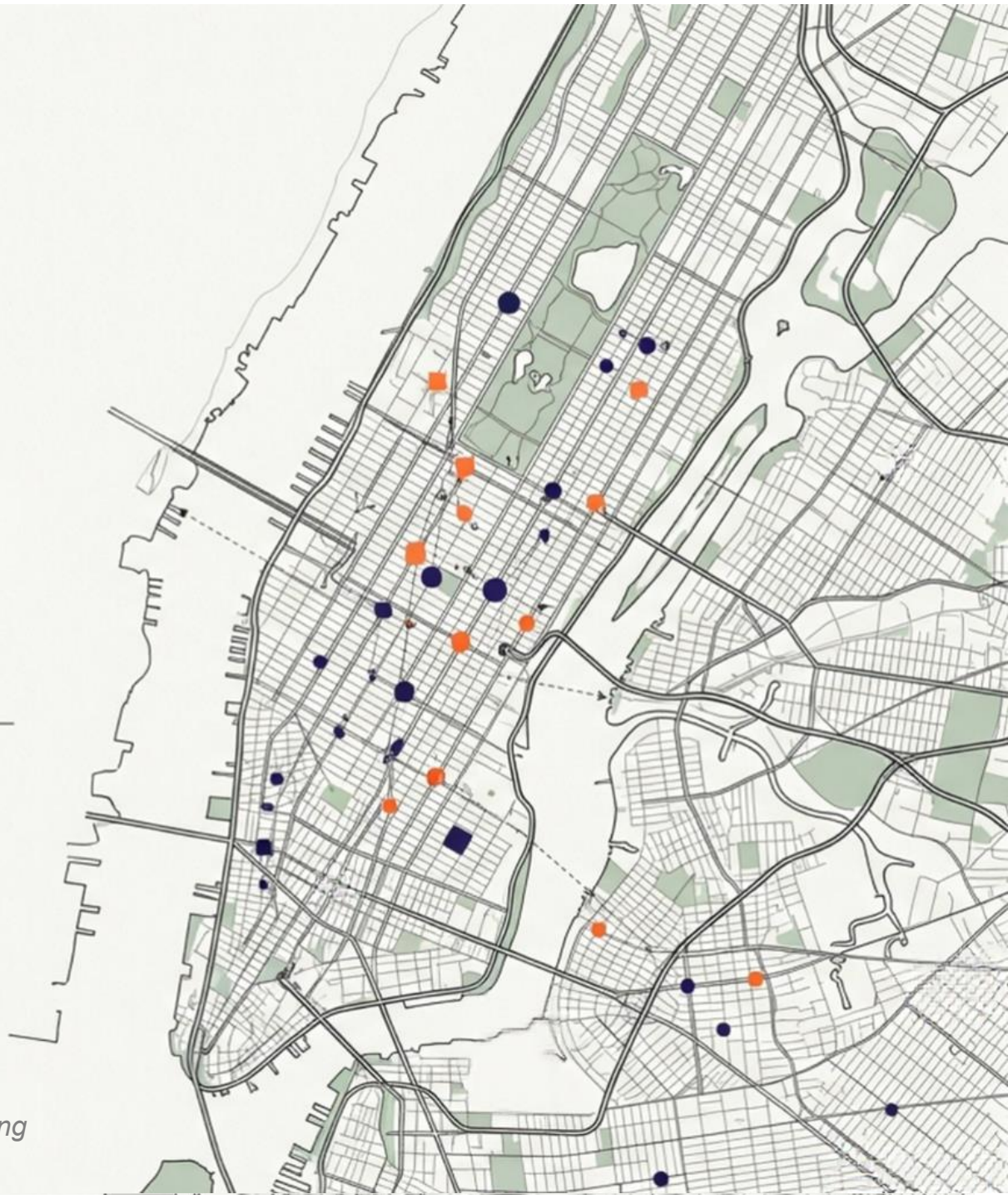
Analyzing Spatiotemporal Patterns of Crime in New York City

BIS 527 Project Proposal

Team | DatumDance

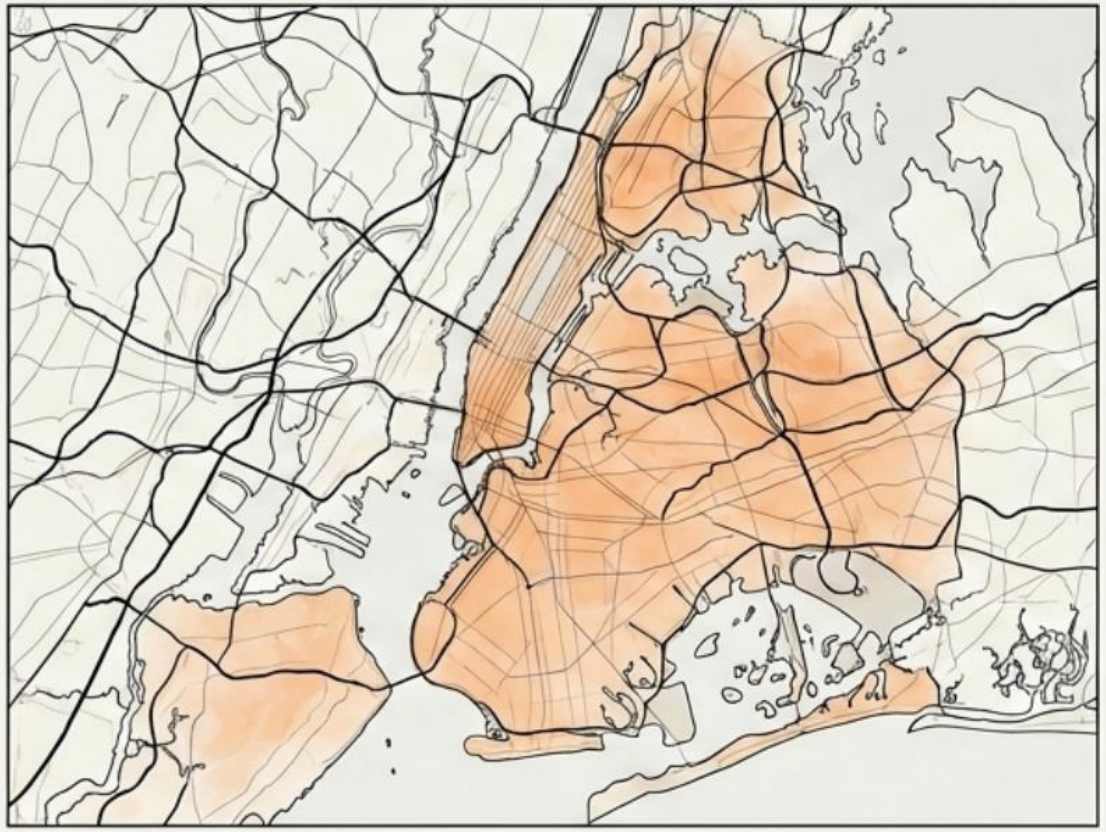
NetID	Full name
jf2546	Ji Feng
jz999	Jiayi Zhu
kz365	Keer Zheng
lq65	Sunny Qin
ml3432	Mengyi Li

AI Use Declaration: The authors used Gemini Notebook only for language polishing and presentation refinement.



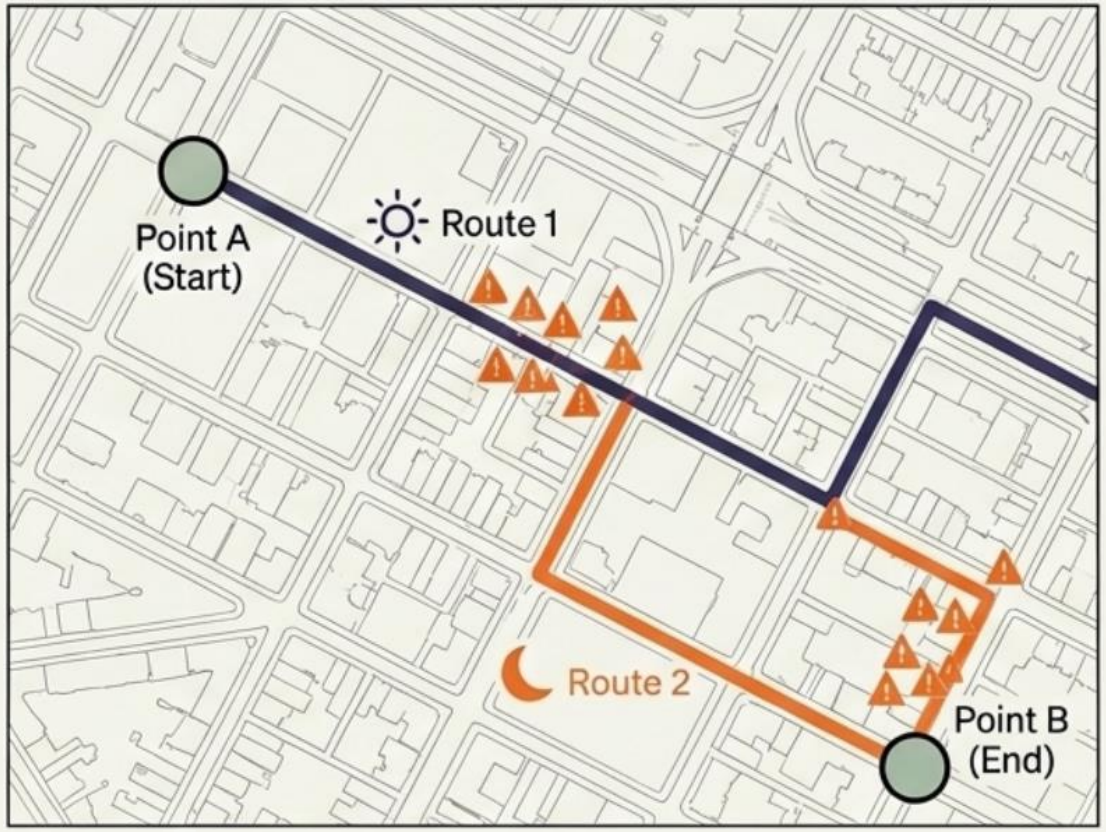
The Limitations of City-Wide Statistics

The Macro View



The Macro View: City-wide stats give a general idea but don't show safety changes in specific places and times.

The Micro Reality - The Two Routes Paradox



The Micro Reality: Two routes between the same start and end points can have very different crime patterns depending on the time and day.

Decoding Spatiotemporal Variance to Profile Route-Level Exposure





The Core Questions

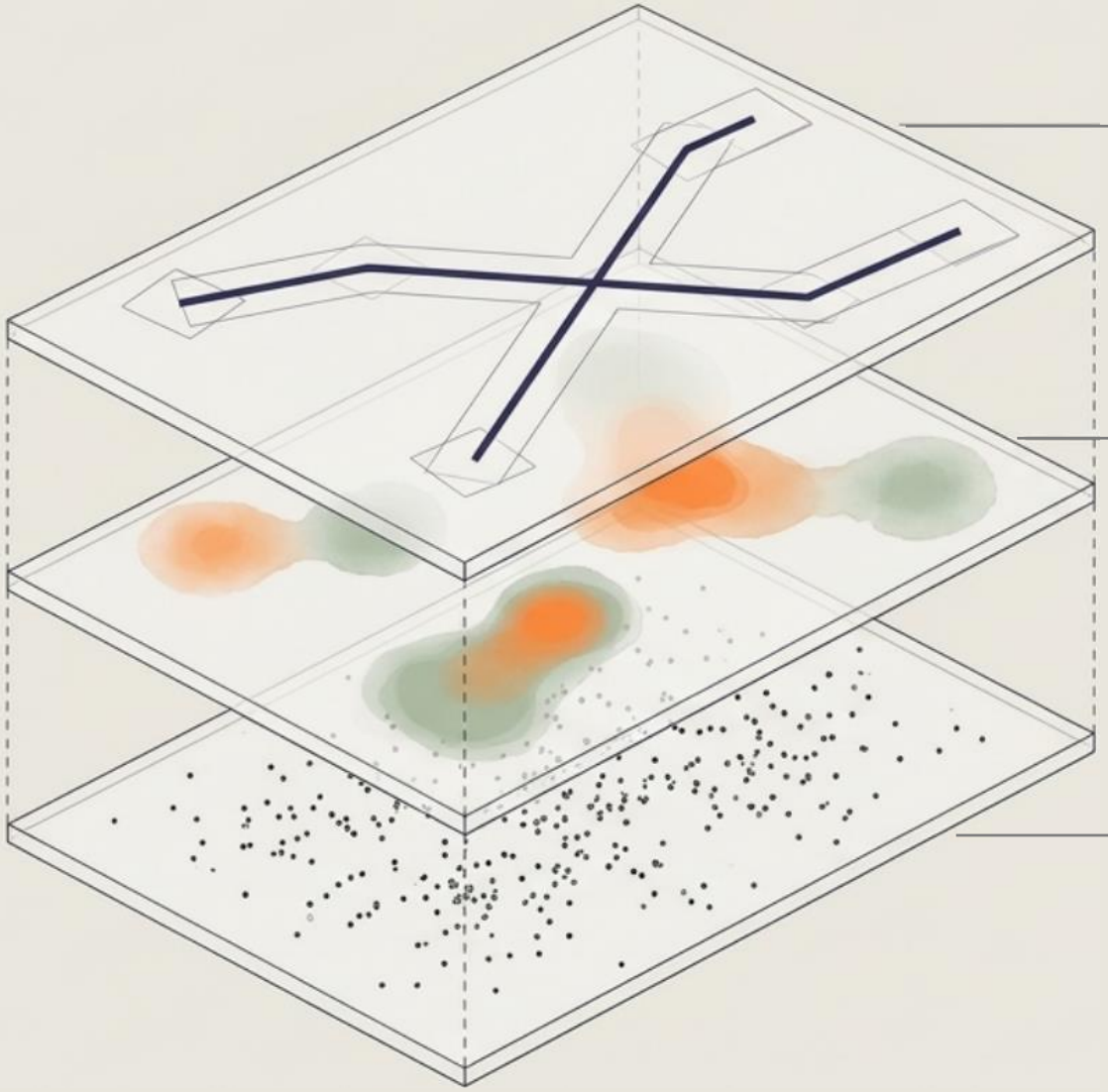
- **Question 1**: How do reported crime categories vary across NYC by location, time of day, and day of the week?
- **Question 2:** How can these shifting macro patterns be utilized to profile historical crime exposure along specific micro-travel routes?

The Goal and Boundary

Project Goal: To engineer an interactive mapping tool enabling users to directly compare historical exposure patterns across alternative routes and shifting travel times.

Analytical Disclaimer: This project strictly describes historical reporting patterns, not predicting an individual person's actual risk.

A Three-Tiered Analytical Pipeline



Layer 3: Route-Level Analysis

Action: The core innovation layer.
Details: Comparing alternative routes with identical origin/destination points through spatial buffers.

Layer 2: Spatiotemporal Patterning

Action: Generating borough-level heatmaps.
Details: Using descriptive statistics, heatmaps, and regression analysis to compare reported crime patterns across crime categories, boroughs, times of day, and days of the week.

Layer 1: Data Foundation

Action: Ingesting and cleaning historical records.
Details: Resolving missing/duplicate records; filtering by critical variables (Date, Time, Location, Crime Category).

Project Milestones and Risk Mitigation Strategy



Risk & Mitigation Protocol

- **Identified Risk:**
  - The quality of the time and location data.
  - The choice of offense categories.

- **Mitigation Strategy:**
  - Dynamic aggregation. If the granular data prove unreliable, the analysis will be adjusted to broader temporal windows or expanded geographic units in order to preserve statistical significance.

Data Architecture and Foundational Frameworks

The Data Engine

NYPD

NYC Open Data

Primary Source: NYPD Complaint Data Historic (via NYC Open Data API, with CSV fallback).

Key Variables Mined



Geospatial Coordinates | **Date/Time** | **Crime Category** | **District**

Theoretical Frameworks

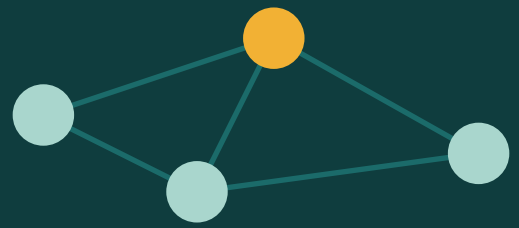
Lim and Li Framework: Establishes the methodology for measuring whole-route exposure rather than relying solely on origin and destination nodes.
doi.org/10.1016/j.tbs.2026.101330

ESCORT Framework (Zhou et al.): Validates the utilization of spatiotemporal patterns to build road-level indices, bridging the gap from city-wide data to micro-route comparisons.
doi.org/10.1109/JSYST.2020.3023762

PRESENTATION 6 OF 11

O(n log n)

Next: DataPulse



Does maternal syphilis spread across neighboring municipalities?

A network analysis of gestational and congenital syphilis in Brazil, 2013–2024

Team O(nlogn): Christine Bi, Cici Wang, Regina Fan, Ruilin Ma, Parash Sapkota

BIS 527 · Introduction to Health Data Science · Fall 2026 · Project proposal

Why it matters?

Congenital syphilis is almost entirely preventable. A mother tested and treated early rarely passes it to her baby.

8.9 → 35.4

syphilis in pregnancy per 1,000 live births, 2014 to 2024 (about 4×)

5.5 → 10.4

congenital syphilis per 1,000, peaking in 2022 and only slightly lower since

24,443

congenital cases reported in Brazil in 2024

If risk moves between connected municipalities, screening can target places whose neighbors are rising, not only places that are already high.

Research Questions

1

Do neighbors matter?

Does a municipality's rate this year depend on its neighbors' rates last year, after accounting for its own past rate and state differences in testing and reporting?

2

Who counts as a neighbor?

A municipality that shares a border, or one that is simply nearby within a fixed distance?

3

Did Sífilis Não work?

Did the 2018 program, which sent support staff to 100 priority municipalities, reduce congenital syphilis, and did the benefit reach their neighbors?

We also map the ratio of congenital to gestational cases (27.2% nationally in 2024). A high ratio points to mothers who were missed or treated too late.

Data Sources

SINAN
Confirmed gestational and congenital cases by mother's municipality of residence and year

SINASC
Live births by municipality, used as rate denominators

IBGE boundaries (geobr)
Municipal borders define neighbors; health regions for robustness checks

Sífilis Não list
Ministry of Health list of the 100 priority municipalities

~67,000

municipality-year rows

5,570 municipalities × 2013–2024. All free public data from DATASUS and IBGE.

First check: our 2024 totals must match the Ministry's 89,724 gestational and 24,443 congenital cases.

Methodology

Spillover model

- Baseline: own rate last year, plus state-by-year intercepts for testing and reporting
- Main model: add neighbors' average rate last year
- Negative binomial counts with live births as offset; repeat on ~450 health regions
- Compare neighbors defined by shared border vs. fixed distance

Did Sífilis Não work?

- Difference-in-differences: 100 priority municipalities vs. similar unselected ones, before and after 2018
- If time allows: untreated neighbors of priority municipalities vs. other untreated ones, to see if the effect spread

Methodology

1

Predict ahead

Fit on 2013–2021, predict 2022–2024: does the neighbor term lower error?

2

Placebo networks

Give each municipality random neighbors; a real effect should vanish.

3

Pre-2018 trends

Priority and comparison municipalities should trend alike before 2018.

Main uncertainty

A neighbor effect could mean infection spreads, or only that nearby places share clinics, labs, and reporting habits.

Priority municipalities were large and high-burden, and a drop in cases could reflect changes in recording, not fewer infections.

Timeline and next steps

Oct 9

Confirm access

Load SINAN, SINASC, and boundaries; match priority list; check 2024 totals

Oct 16

Baseline analysis

Build border and distance networks; map rates and ratio; fit baseline model

Oct 19–23

Mid-point check-in

Add neighbor term; run placebos and prediction check; choose comparison group

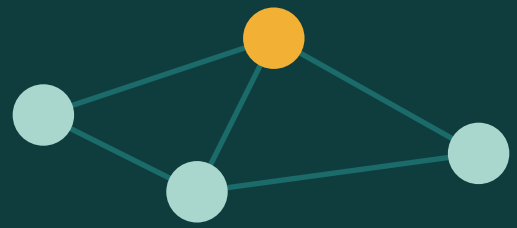
Nov 10–20

Final results

Freeze results Nov 10; poster in week 12; final report in finals week

First step: set up a shared repository, match the priority list to IBGE codes, and download one year of SINAN data to confirm it records the mother's municipality of residence.

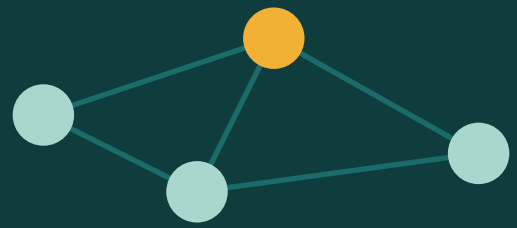
Reference



Brasil. Ministério da Saúde. Secretaria de Vigilância em Saúde e Ambiente. (2025). Boletim Epidemiológico: Sífilis 2025. Número Especial, outubro de 2025. Brasília, DF: Ministério da Saúde.

Claude (Anthropic) was used to search for datasets and the Sifilis Nao municipality list, check the national figures, and suggest analysis methods.

We opened the TabNet and archived list links ourselves. Not yet checked: the case and birth data, which are not downloaded yet, and whether 2024 data are final.



Thank You

PRESENTATION 7 OF 11

DataPulse

Next: MindMerge

Fertility and maternal mortality data analysis

Team name

DataPulse

Thursday, October 1, 2026

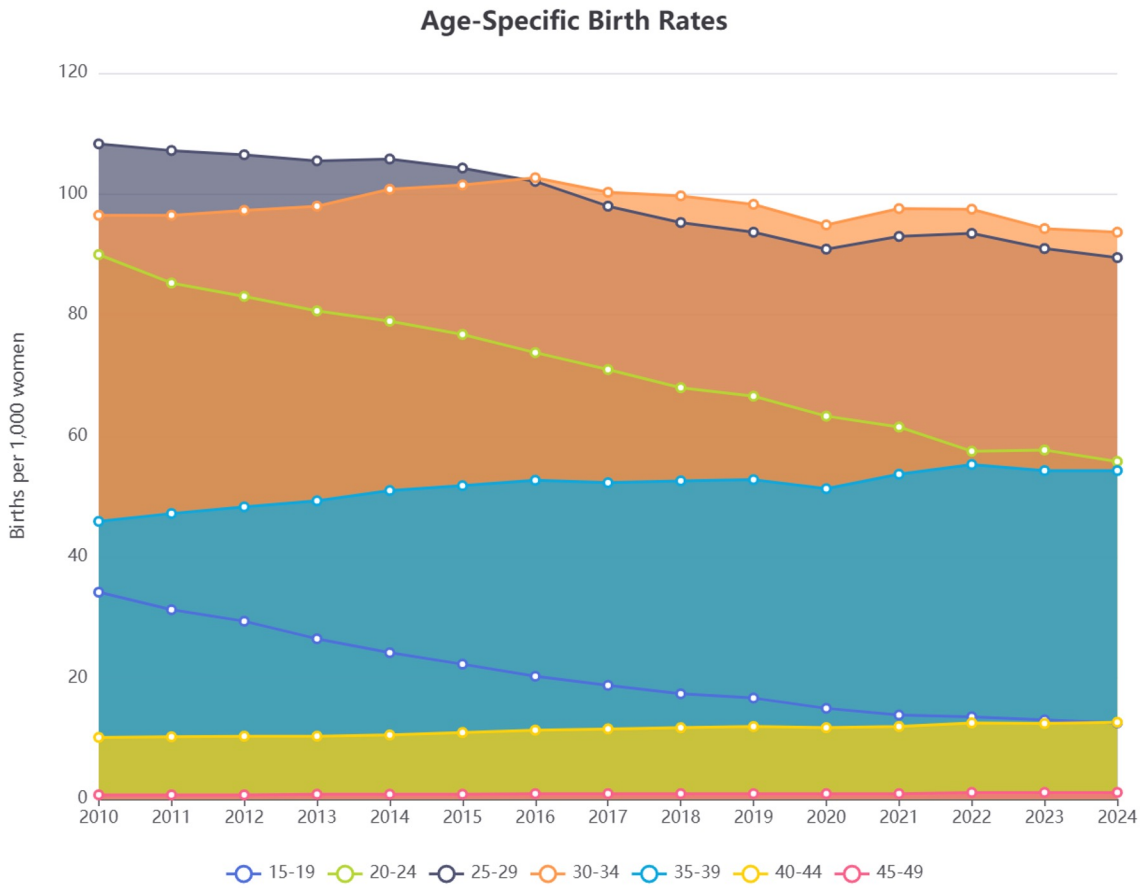
Team members

NetID	Full name
jl4593	Joanna Li
yc2337	Cindy Chen
yz2574	Tristan Zhong
yz2598	Yuchen Zhang
zs452	Zhentao Shen

Background and main question

Background

In recent years, fertility patterns in the United States have shifted toward older maternal ages.

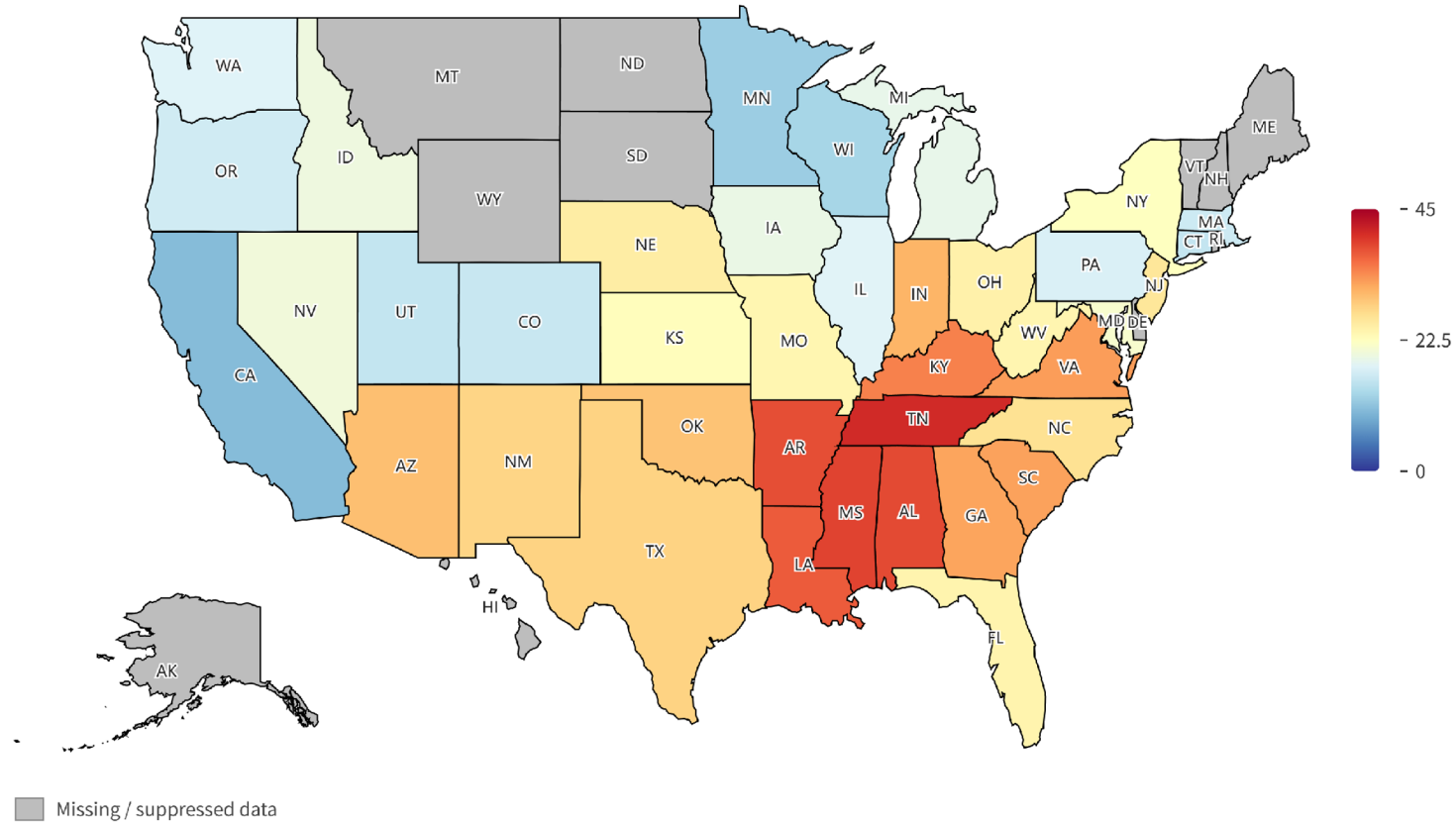


Main question

Our project will examine maternal mortality among U.S. women of reproductive age. We aim to compare mortality across age groups and identify demographic and health-related factors associated with maternal death.

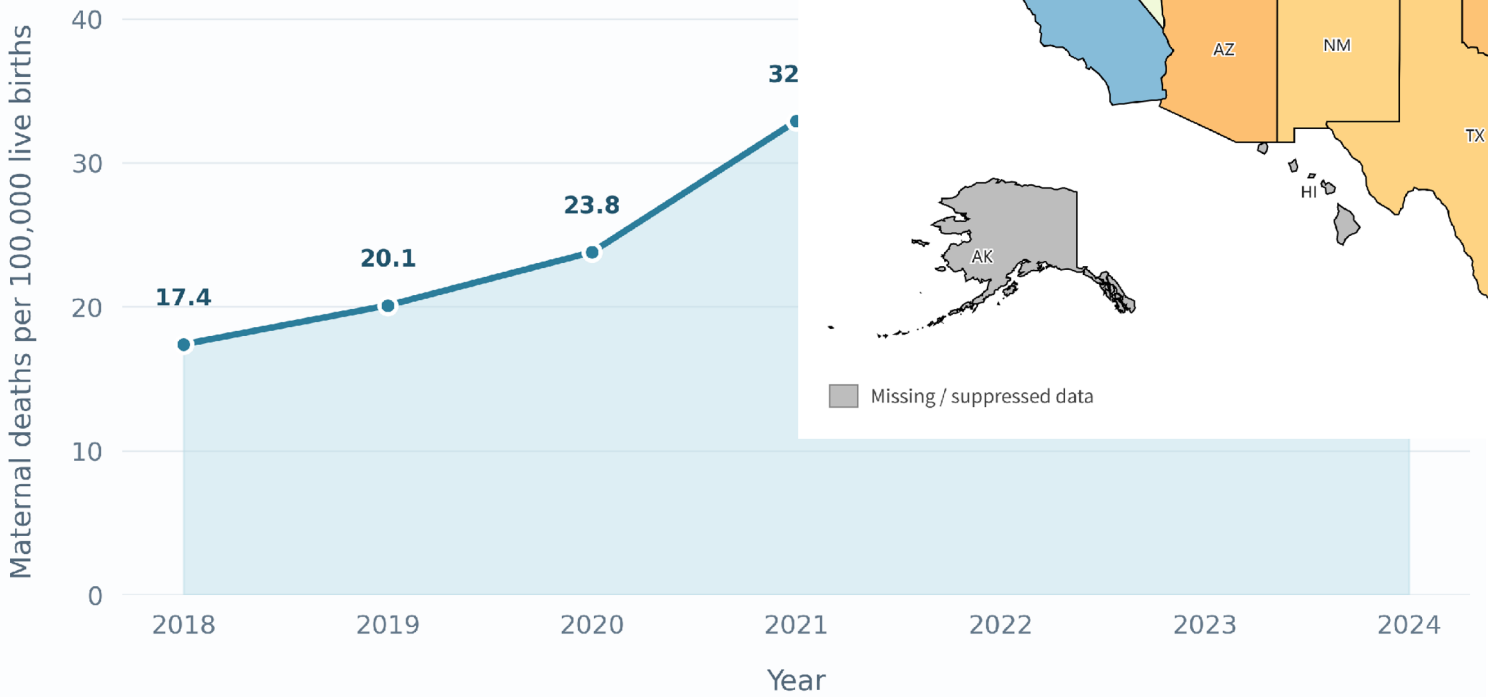
U.S. Maternal Mortality by State (2018–2022)

Maternal deaths per 100,000 live births | Pooled period
Source: NCHS / NVSS | U.S. total: 23.2



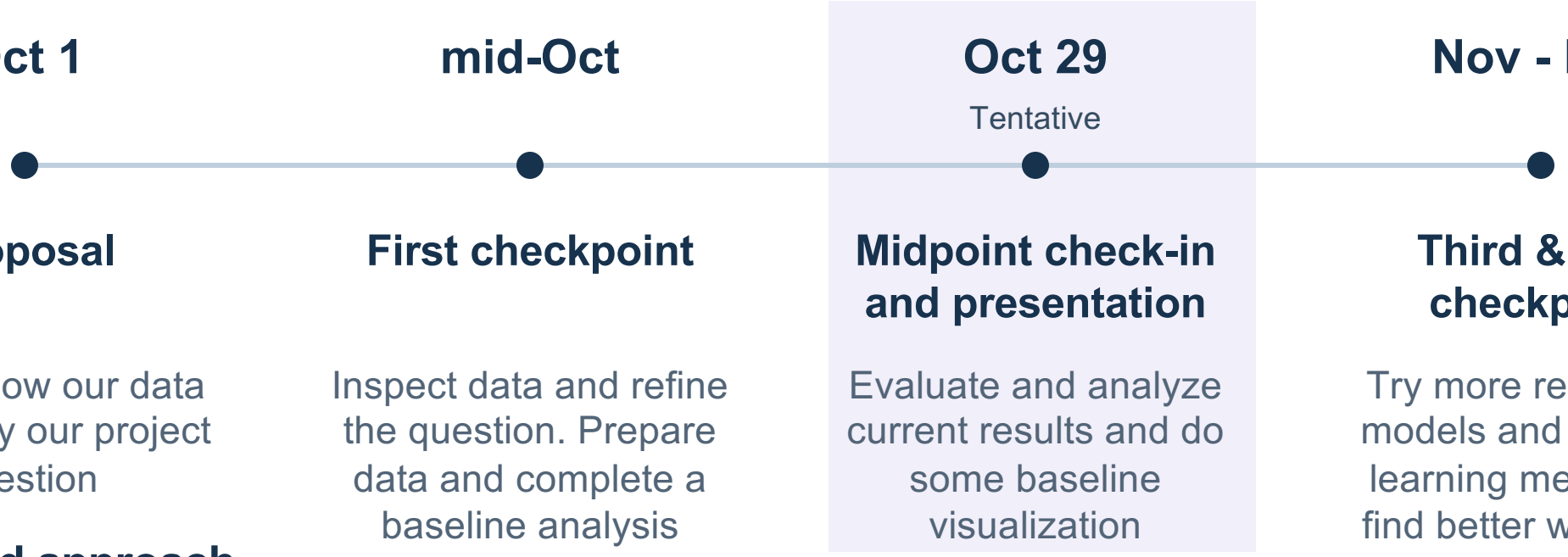
National Maternal I

2018–2024 | All races and



Project path and timeline

Tentative dates and expected outputs

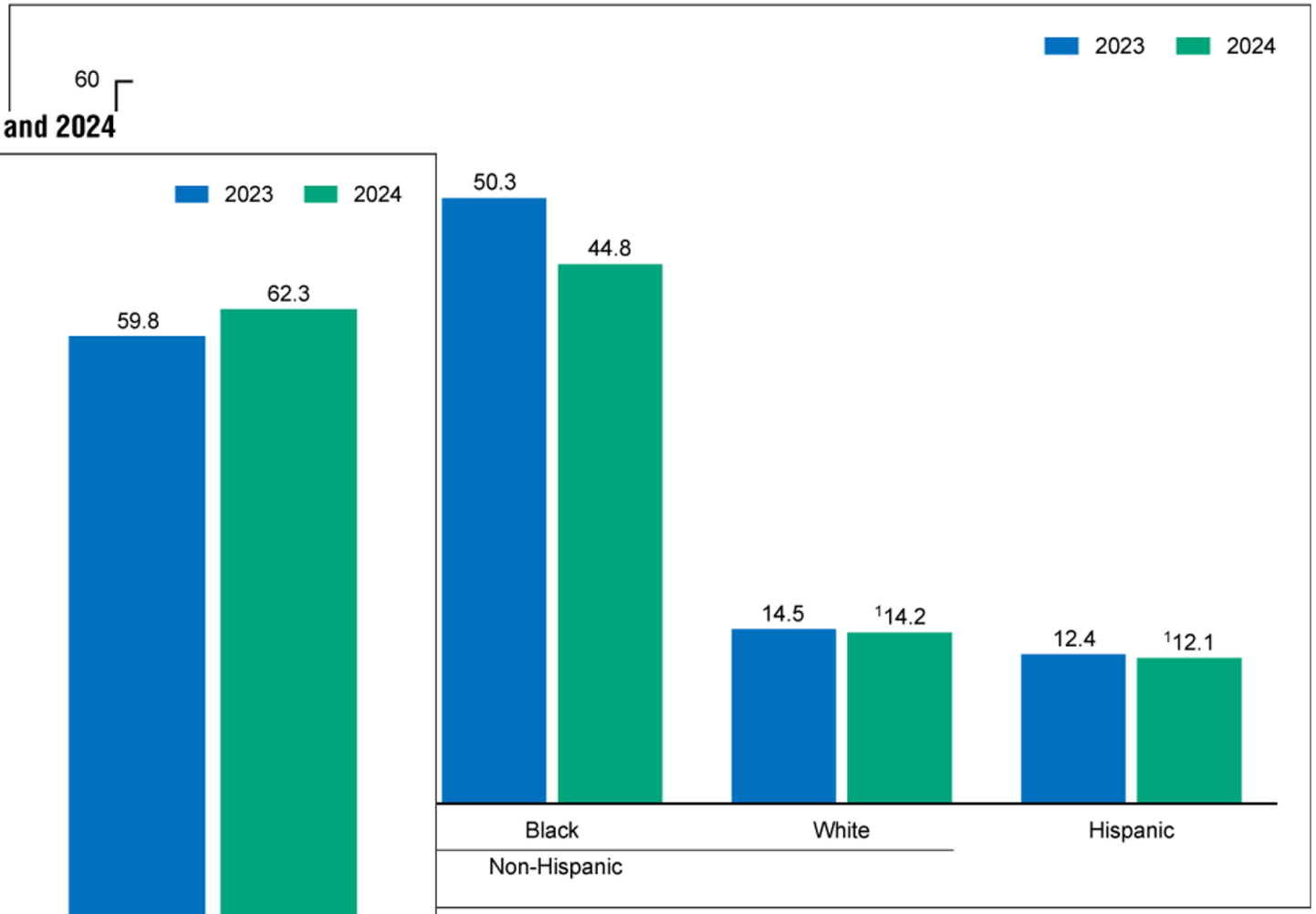


Proposed approach

The baseline analysis will be primarily descriptive. The maternal mortality rates will be summarized by factors including year and age group. To provide context for changes in maternal age, trends can be visualized in age-specific birth rates and mean maternal age from 2010–2024. Analyses directly comparing fertility and maternal mortality trends will focus on the overlapping 2018–2024 period.

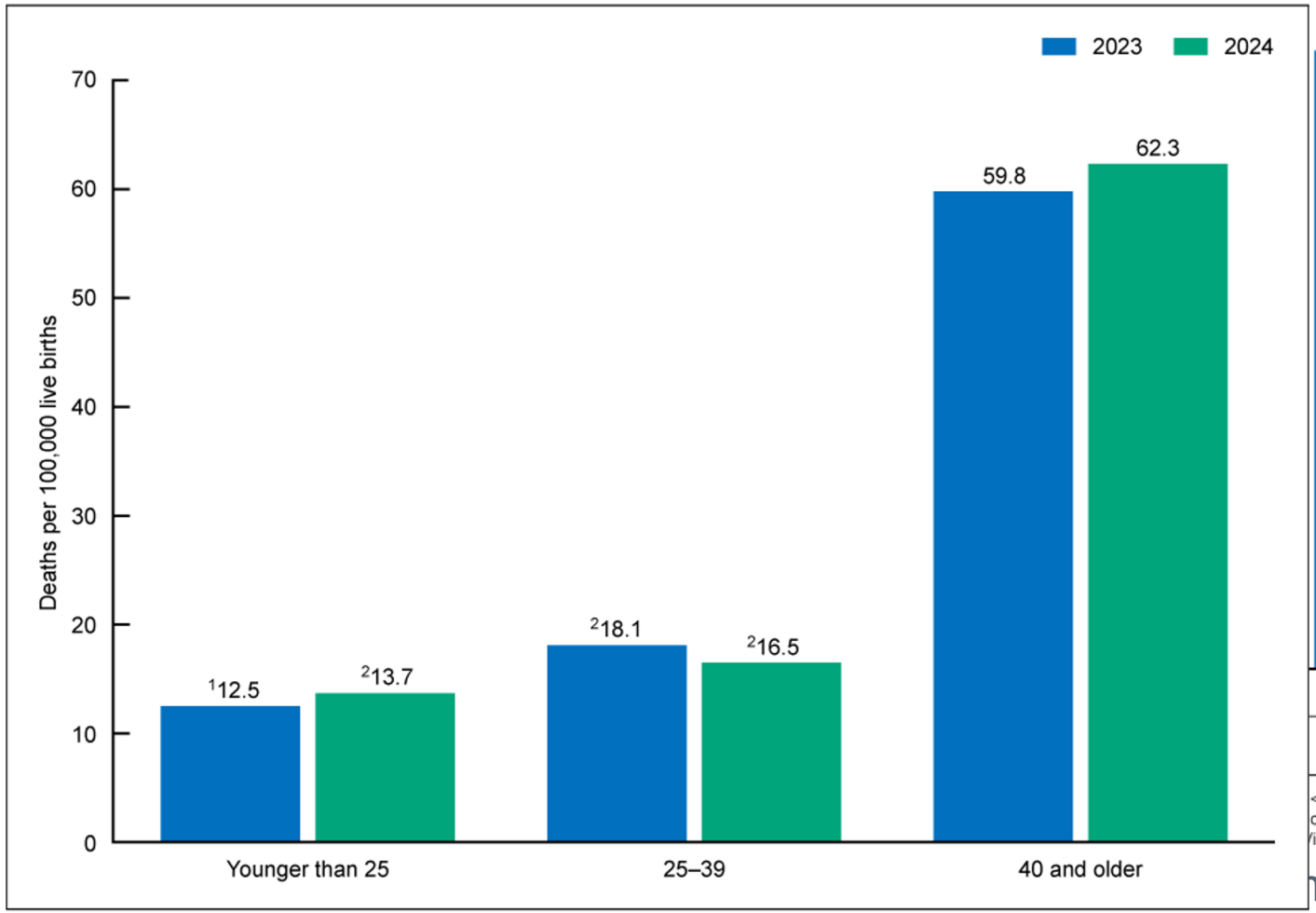
Resources

Figure 2. Maternal mortality rate, by race and Hispanic origin: United States, 2023 and 2024



< 0.05).
origin may be of any race.
ital Statistics System, mortality data file.

Figure 3. Maternal mortality rate, by age group: United States, 2023 and 2024



[1]Significantly different from ages 25–39 and 40 and older ($p < 0.05$).
[2]Significantly different from age 40 and older ($p < 0.05$).
SOURCE: National Center for Health Statistics, National Vital Statistics System, mortality data file.

hay be used to assist some code in data

PRESENTATION 8 OF 11

MindMerge

Next: Pierpont

Long-Term PM2.5 Exposure and Adult Asthma Prevalence Across U.S. Counties

Team name

MindMerge

Thursday, October 1, 2026

Team members

NetID	Full name
jc4693	Jie Cang
jh3637	Jiahui He
sc3648	Siyu Chen
xw558	Xinyu Wang
yz2584	Yuanyumeng Zhu

Background and main question

Background

PM2.5 is an important air pollutant linked to respiratory health, and asthma remains a common chronic respiratory condition. Prior studies have examined either single-year PM2.5 exposure or broader asthma determinants. The long-term county-level association between PM2.5 and asthma prevalence remains unclear.

Main question

Across U.S. counties, is higher long-term PM2.5 exposure associated with higher age-adjusted adult current asthma prevalence, after accounting for smoking, lack of health insurance, and obesity?

Project path and timeline

Tentative dates and expected outputs

Oct 1

Proposal

Set the study question, data sources, and analytic plan.
Present.

.

Oct 16

Data + first analysis

Standardize 5-digit FIPS.
Document unmatched/missing.
Compute mean 2017–2021 PM2.5.
Merge with 2022 PLACES.
Plot PM2.5 vs. asthma.

Oct 29

Tentative

Midpoint check-in and presentation

Show descriptives.
Check distributions.
Fit simple and nested linear regressions.

Dec 1

Final analysis

Check residual, Q–Q, influence plots, and VIF.
Run PM2.5 sensitivity analyses.
Finalize poster & report.

Proposed approach

- Use nested county-level linear regressions to estimate the PM2.5–asthma association, adjusting for three covariates.
- Quantity of interest: the PM2.5 coefficient with a 95% CI.
- Main uncertainty: county geography/FIPS matching. We will inspect unmatched and missing records before analysis.

$$Asthma_i = \beta_0 + \beta_1 PM2.5_i + \beta_2 Smoking_i + \beta_3 Uninsured_i + \beta_4 Obesity_i + \varepsilon_i$$

Resources

Datasets

1. CDC Daily County-Level PM2.5 Concentrations
2. CDC PLACES County Data, 2024 release

PM2.5: 2017–2021 population-weighted daily estimates (contiguous U.S.)
PLACES: 2022 county-level asthma, smoking, uninsured, and obesity estimates

CDC PM2.5 | CDC PLACES

Access

Publicly downloadable; no restricted-data application.

Relevant Literature

[1] Young et al. (2014)
Ambient air pollution and incident adult asthma in U.S. women
doi:10.1164/rccm.201403-0525OC

Motivates studying PM2.5 exposure in relation to adult asthma.

[2] Lotfata et al. (2023)
U.S. county-level asthma prevalence determinants; a cross-sectional study

doi:10.1186/s12942-023-00343-6

Supports a county-level approach and adjustment for contextual health factors.

Other resources

R for cleaning, visualization, and regression; CDC documentation and PLACES release notes.

PRESENTATION 9 OF 11

Pierpont

Next: jet2holiday

Predicting Public Health Impact from Air Quality and Weather Conditions

Team name

Pierpont

Thursday, October 1, 2026

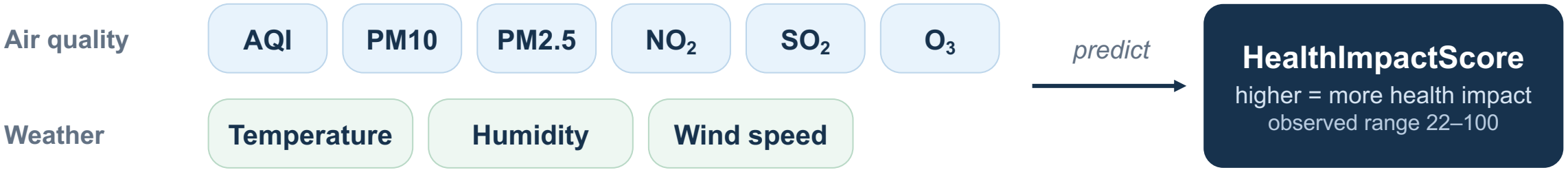
Team members

NetID	Full name
jw3362	Jialei Wu
rh2264	Ruiqi Hu
sw2592	Shuohang Wu
yc2364	Yeran Chen
tc994	Tin Long Chung

Background and main question

Background

Air pollution harms health, and its levels move with the weather.



Population: 5,811 records in the Kaggle dataset · **Observational:** associations, not cause and effect

Main question

How well can air-quality and weather data predict the score, and which factors are most strongly linked to it?

Project path and timeline

Tentative dates, expected outputs and owners

Oct 1

Proposal

Plan + feedback

Whole team

Early–mid Oct

Explore data, first regressions

Score check + plots

Ruiqi, Jialei, Shuohang

Oct 29

Tentative

Midpoint check-in and presentation

Held-out RMSE and R^2

Yeran, Tin Long

Early Dec

TBC

Poster and final report

Final models + figures

Whole team

Proposed approach

1 Baseline
AQI only

→

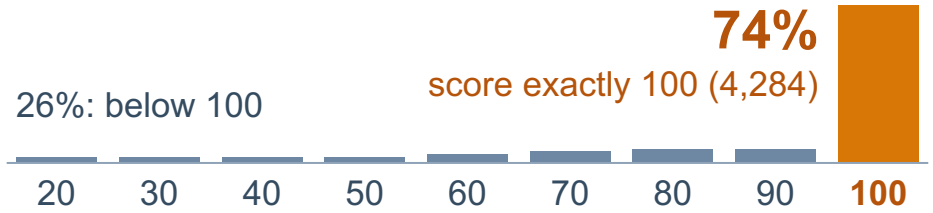
2 Full regression
all 9 variables

→

3 Tree / forest
only if nonlinear

How well? held-out RMSE and R^2
Which factors? standardized coefficients

Main obstacle: a ceiling at 100



Next: check how the score is built
Backup to consider: hospital-admission counts

Resources

Dataset(s)

Air Quality and Health Impact Dataset

Kaggle · 5,811 records; 9 predictors and the score
Region and time period not documented

kaggle.com/datasets/rabieelkharoua/air-quality-and-health-impact-dataset

Access

✓ Public CSV (CC BY 4.0), already downloaded
✓ No missing values
→ Results apply to this dataset only

Relevant Literature

Dockery et al. (1993). Air pollution and mortality in six U.S. cities. NEJM

doi.org/10.1056/NEJM199312093292401

→ Classic evidence that fine particles track mortality; why PM2.5 and PM10 are in our model

Di et al. (2017). Air pollution and mortality in the Medicare population. NEJM

doi.org/10.1056/NEJMoa1702747

→ PM2.5 and ozone linked to deaths even below the standards of the time: model jointly, check for curves

Other resources

R or Python (regression, random forest) · a laptop is enough · no special permissions needed

PRESENTATION 10 OF 11

jet2holiday

Next: M4zi1

Factors Associated with Future Fertility Intentions among U.S. Women Aged 20–49

Team name

Jet2Holiday

Thursday, October 1, 2026

Team members

Sort alphanumerically by NetID.

NetID	Full name
bd594	Baichuan Duan
rw796	Harrison Wang
sw2584	Sujun Wu
yd454	Yijing Duan
yz2576	Yijia Zhang

Background and main question

Background

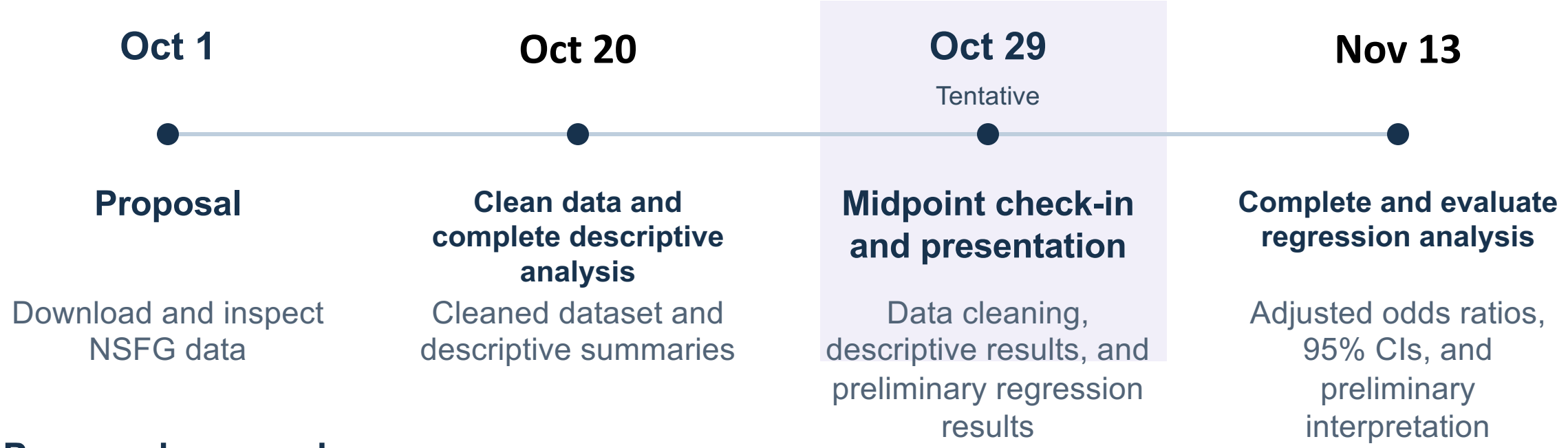
Fertility intentions reflect individuals' plans regarding future childbearing. Understanding how fertility intentions vary across demographic and socioeconomic groups can provide insights into reproductive decision-making and public health.

Main question

Among women aged 20–49 in the United States, how are age, socioeconomic factors, relationship status, and past reproductive history associated with future fertility intentions?

Project path and timeline

Tentative dates and expected outputs



Proposed approach

Fit a multivariable logistic regression with fertility intention (INTENT) as the binary outcome, accounting for the NSFG complex survey design. Evaluate estimates using adjusted odds ratios, 95% CIs, and model adequacy (e.g., linearity of age). Main obstacle: missing data and variable coding.

Resources

Dataset(s)

2022–2023 National Survey of Family Growth (NSFG), Female Respondent File — NCHS/CDC

U.S. women aged 20–49; INTENT, AGER, HIEDUC, TOTINCR, LABORFOR, RMARITAL, PARITY

CDC/NCHS NSFG 2022–2023 Public-Use Data

Access

Public-use data; available for download from NCHS/CDC. No Research Data Center access expected.

Relevant Literature

Morales (2020), “Partners’ Educational Pairings and Fertility Intentions in the United States”

DOI: 10.1177/2329496520906515

Uses 2015–2017 NSFG to show how partners’ educational pairings relate to women’s intentions for additional children.

Shreffler et al. (2016), “Infertility and Fertility Intentions, Desires, and Outcomes among US Women”

DOI: 10.4054/DemRes.2016.35.39

Uses NSFB data to show how infertility and parity are associated with fertility intentions among U.S. women.

Other resources

Python/R for data cleaning, survey-weighted analysis, and visualization; NSFG codebook and survey design documentation.

PRESENTATION 11 OF 11

M4zi1

Shifting Payer Spending — The Effects of Medicaid Expansion

Team name

M4Zi1

Thursday, October 1, 2026

Team members

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[yc2375]	[Yunqing Chen]
[yl2998]	[Yiming Lu]

Background and main question

Background

The ACA Medicaid expansion broadened Medicaid eligibility for low-income adults and substantially increased insurance coverage. Previous studies have documented effects on access to care and health care utilization.

Main question

How did Medicaid expansion change health care spending across different payers in U.S. states?

Project path and timeline

Tentative dates and expected outputs



Proposed approach

We will use a difference-in-differences model to estimate spending changes after Medicaid expansion and evaluate it by checking pre-expansion trends and comparing effects across payers.

Resources

Datasets

United States Healthcare Spending by Health Condition and County 2010-2019 — DEX2.0
Population: U.S. population across 3,110 counties
Variables: per-capita spending, payer, and type of care.
https://ghdx.healthdata.org/about-ghdx

Status of State Action on the Medicaid Expansion Decision – KFF
U.S. states. Reference date: August 2026
Variable: Medicaid expansion implementation timing
https://www.kff.org/affordable-care-act/state-indicator/state-activity-around-expanding-medicaid-under-the-affordable-care-act/

Current Population Survey — U.S. Census Bureau & BLS
U.S. population. 1994-2020
Variables: unemployment, income, poverty
https://www.census.gov/programs-surveys/cps.html

ACS 1-Year Estimates — U.S. Census Bureau
State-level U.S. population, 2005-2024
Variables: unemployment, income, poverty
https://www.census.gov/data/developers/data-sets/acs-1year.html

Relevant Literature

Paper 1: Mazurenko et al., 2018,
The Effects Of Medicaid Expansion Under The ACA
Provides prior evidence on expansion outcomes and helps position our project as an extension focused specifically on payer-specific health spending.
doi: https://doi.org/10.1377/hlthaff.2017.1491

Paper 2: Sommers et al., 2017
Three-Year Impacts Of The Affordable Care Act
Shows that Medicaid expansion reduced out-of-pocket spending and improved access to care.
doi: https://doi.org/10.1377/hlthaff.2017.0293

Access: All publicly available
Other resources: ChatGPT